B. Tech. Computer Science and Engineering (Specialization in Artificial Intelligence & Machine Learning)

Semester – V

Sr. No.	Course Type	Course Code	Course Name	Teaching Scheme (Hrs./Week)				Examination Scheme				Total Marks
				L	T	P	C	Formative Assessment CIA		Summative Assessment ESE		
								Theory	Lab	Theory	Lab	
1	PCC	NYCM501	Design and Analysis of Algorithm	3	--	--	3	50	--	100	--	100
2	PCC	NYCM502	Operating System	3	--	--	3	50	--	100	--	100
3	PCC	NYCM503	Computer Vision	3	--	--	3	50	--	100	--	100
4	PCC	NYCM511	Operating System Laboratory	--	--	4	2	--	50	--	50	100
5	PCC	NYCM512	Computer Vision Laboratory	--	--	2	1	--	25	--	25	50
6	PEC	NYCME_	Program Elective-I	3	--	--	3	50	--	100	--	100
7	OE		Open Elective III	2	--	--	2	50	--	100	--	100
8	MDM		Minor course 3 #	3	--	--	3	50	--	100	--	100
9	AEC (HSSM)	NHSA13	Essential Aptitude Skills	--	--	4	2	--	50	--	50*	100
10	VAC		Tableau and Power BI	--	--	2	--	--	--	--	--	--
TOTAL				**17**	**--**	**12**	**22**	**300**	**125**	**600**	**125**	**850**

Year: Third Year **Semester – V**
Course: Design and Analysis of Algorithm **Course Code: NYCM501**

Teaching Scheme (Hrs. /Week)				Continuous Internal Assessment (CIA)					End Semester Examination		Total
L	T	P	C	CIA-1	CIA-2	CIA-3	CIA-4	Lab	Theory	Lab	
3	0	0	3	10	20	10	10	--	50	--	100
Max. Time, End Semester Exam (Theory) - 3Hrs.									End Semester Exam (Lab) - 0Hrs.		

Prerequisites:
Data Structures And Algorithm, Basic Mathematics

Objectives:	
Students are able to:-	
1	**Analyze Efficiency:** To master asymptotic notations and mathematical techniques (like Master's Theorem) for evaluating the time and space complexity of AI-scale algorithms.
2	**Strategic Design:** To understand and apply diverse algorithmic strategies—such as Greedy, Dynamic Programming, and Backtracking—to solve complex optimization problems.
3	**Graph & Network Proficiency:** To leverage graph theory and traversal techniques for modeling state-space searches, neural network topologies, and relationship mapping.
4	**Manage Intractability:** To distinguish between solvable (P) and hard (NP) problems and learn to apply approximation and randomized techniques when an exact solution is computationally impossible.
5	**Modern Computational Literacy:** To explore advanced frontiers like String Matching for NLP and Quantum Algorithms to understand the future of high-speed computation.

Unit No	Details	Hours
1	**Module 1: Introduction to Algorithms:** Overview and Importance of Algorithms, Characteristics of Algorithm. **Stages of algorithm development:** Describing the problem, Identifying a suitable technique, Design of an algorithm. **Algorithmic Strategies:** Brute-Force, Heuristics and Divide and Conquer, Illustrations of these techniques Problem for Problem-Solving **Module 2: Analysis of Algorithm:** Asymptotic analysis of Complexity Bounds - Best, Average and Worst-Case behavior, Performance Measurements of Algorithm, Time and Space Trade-Offs, Analysis of Recursive Algorithms through Recurrence Relations	9

	Divide and Conquer analysis: Binary Search, Quick Sort, Merge Sort Analysis of all algorithms.	
2	**Module 1: Dynamic Programming:** Dynamic Programming--Elements of Dynamic Programming, Rod Cutting, Matrix chain multiplication, Longest Common **Module 2: Greedy Algorithms-** Activity Selection Problem, Elements of greedy strategy, Knapsack problem, Huffman Coding; Fibonacci Heaps.	9
3	**String Matching Algorithms and Graph Algorithms** Naïve String-matching Algorithms, KMP algorithm, Rabin-Karp Algorithm, **All pair shortest path:** Bellman Ford Algorithm, Floyd-Warshall Algorithm **Network Flows:** Flow Networks, Maximum Flows: Ford-Fulkerson, Edmond-Karp Algorithm – Application of Max Flow to maximum matching problem	9
4	**Geometric Algorithms and Randomized algorithms** **Line Segments:** Properties, Intersection, sweeping lines – **Convex Hull finding algorithms:** Graham's Scan, Jarvis' March Algorithm. **Randomized Algorithms:** The hiring problem - Finding the global Minimum Cut.	9
5	**Classes of Complexity and Approximation Algorithms** The Class P - The Class NP - Reducibility and NP-completeness – SAT (Problem Definition and statement), 3SAT, Independent Set, Clique, Approximation Algorithm – Vertex Cover, Set Cover	9

Outcomes:	
On completion of the course, learner will be able to :-	
CO1:	Analyze the complexity of recursive and non-recursive algorithms using **Asymptotic notations** and **Master's Theorem**.
CO2:	Apply **Dynamic Programming** and **Greedy strategies** to solve optimization problems like Knapsack and Matrix Chain Multiplication.
CO3:	Design and implement **Graph algorithms** (DFS, BFS, MST) and classify problems into **P, NP, NP-Hard, and NP-Complete** categories.
CO4:	Solve combinatorial search problems using **Backtracking** and **Branch and Bound** techniques (e.g. N-Queens, TSP).
CO5:	Evaluate **Randomized, Approximation, and String Matching** algorithms, and explain the fundamental concepts of **Quantum computing**.

Text Books

1. Fundamental of computer algorithms, Horowitz, E., Sahni, S., & Rajasekaran, S. Hyderabad, Universities Press; Second edition, 2008.
2. Design and Analysis of Algorithms, Kleinberg J, Tardos E.. Pearson Education India; 2006

Reference Books
1. 1. Knuth Donald E, "Art of Computer Programming: Fundamental Algorithms Volume 1 Fundamental Algorithms", Third Edition, Pearson Publishers, 2011. 2. Aho A.V., Hopcraft J. E., Dulman J. D., The Design and Analysis of Computer Algorithms, Addsion Wesley (1974) 1st ed. 3. S. K. Basu, "Design Methods and Analysis of Algorithm", PHI. 4. John Kleinberg, Eva Tardos, "Algorithm Design", Pearson.
Online Resources
• https://nptel.ac.in/courses/106101060 • https://nptel.ac.in/courses/106106131 • https://www.tutorialspoint.com/design_and_analysis_of_algorithms/index.htm